

Ergodicity breaking in quantum circuits synthesized from elementary QCA

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Wide range of models, ready for quantum processors, diverse physics, MPO and MPDO based classical computability / entanglement study.

I. CITATION NOTES

- quantum ECA, Goldilocks [1]
- Rule 150 [1–3]
- entanglement studies [4–6]
- Experiments [2, 7]
- error correction [8]
- operator growth, ETH, some models [9, 10]
- scrambling, Clifford QCA [11, 12]
- DKCA, phase transitions [13–15]
- Rule 54 [16–20]
- circuit theory [21, 22]
- application SPT phases [23]
- reversible simulation of irreversible CA [24]
- Rydberg proposal [25]
- Rule 201 [26]
- Rule 156 [27, 28]
- update order [29]
- abstract models and classification [30]

II. INTRODUCTION

Quantum cellular automata (QCA) present a versatile framework for exploring interacting quantum many-body systems, exhibiting highly complex behaviors that emerge from the repeated translation-invariant application of simple local unitary [31], later extended to non-unitary [4], instructions to a chain of qubits. Despite this simplicity, QCA are capable of universal quantum computation [31] and serve as a versatile testbed for investigating non-equilibrium phenomena, including phase transitions, quantum many-body scars, and quantum chaos. The locality and simple instruction sets of QCA present a major advantage for NISQ-era quantum computing with limited controls on individual qubits and their interactions. Recent experiments have successfully

demonstrated the feasibility of QCA in the laboratory, ranging from the simulation of single-particle Dirac quantum walks on trapped-ion [32] and photonic devices [33], entanglement generation on superconducting processors [2] and notably on dual-species Rydberg arrays with global controls only [7]. However, bridging the gap between the exponentially large rule space with its many theoretical model QCA and executable quantum processor instructions remains non-trivial.

In Section III of this Letter, we present a universal circuit synthesis for the set of 256 so-called *elementary* QCA [], mapping both reversible and irreversible rules into minimal quantum circuits via Kronecker Reed-Muller expansions [21, 22]. In Section IV, we then numerically investigate the capabilities of these QCA to generate entanglement beyond the limits of classical simulability using Matrix Product States (MPS) [34]. We find that unitary QCA can produce both area-law or volume-law entanglement depending on the initial state, which is due to different kinds of ergodicity breaking, including low-entanglement subsectors and isolated quantum many-body scars. To bypass the initial state dependence, we use a Matrix Product Operator (MPO) encoding of the single-cycle unitary time evolution operator to assert a rule’s inherent complexity. We uncover non-trivial effective Schmidt rank scalings, including linear growth due to strong Hilbert space fragmentation. Finally, we extend the MPO-based analysis to superoperators obtained from non-unitary QCA circuits in ??.

III. CIRCUIT SYNTHESIS

Consider a chain of N classical bits in state 0 or 1. At each discrete time step, the sites are updated according to a local transition function f , which depends only on the states of the bit itself and its two direct neighbors. Because the domain of f has $2^3 = 8$ possible configurations, there are $2^8 = 256$ distinct rules. These are canonically indexed as C_R using the Wolfram rule number, which interprets the binary 8-bit output vector of the truth table as a decimal integer $R = \sum_{k=0}^7 c_k 2^k$, see Fig. 2 a). This class of simple, so-called *elementary* cellular automata (ECA) is known to exhibit rich dynamical classical behaviour, including fixed points, periodic orbits, chaotic pseudo-random number generators, and universal Turing computability [].

We translate the full set of 256 ECAs into quantum

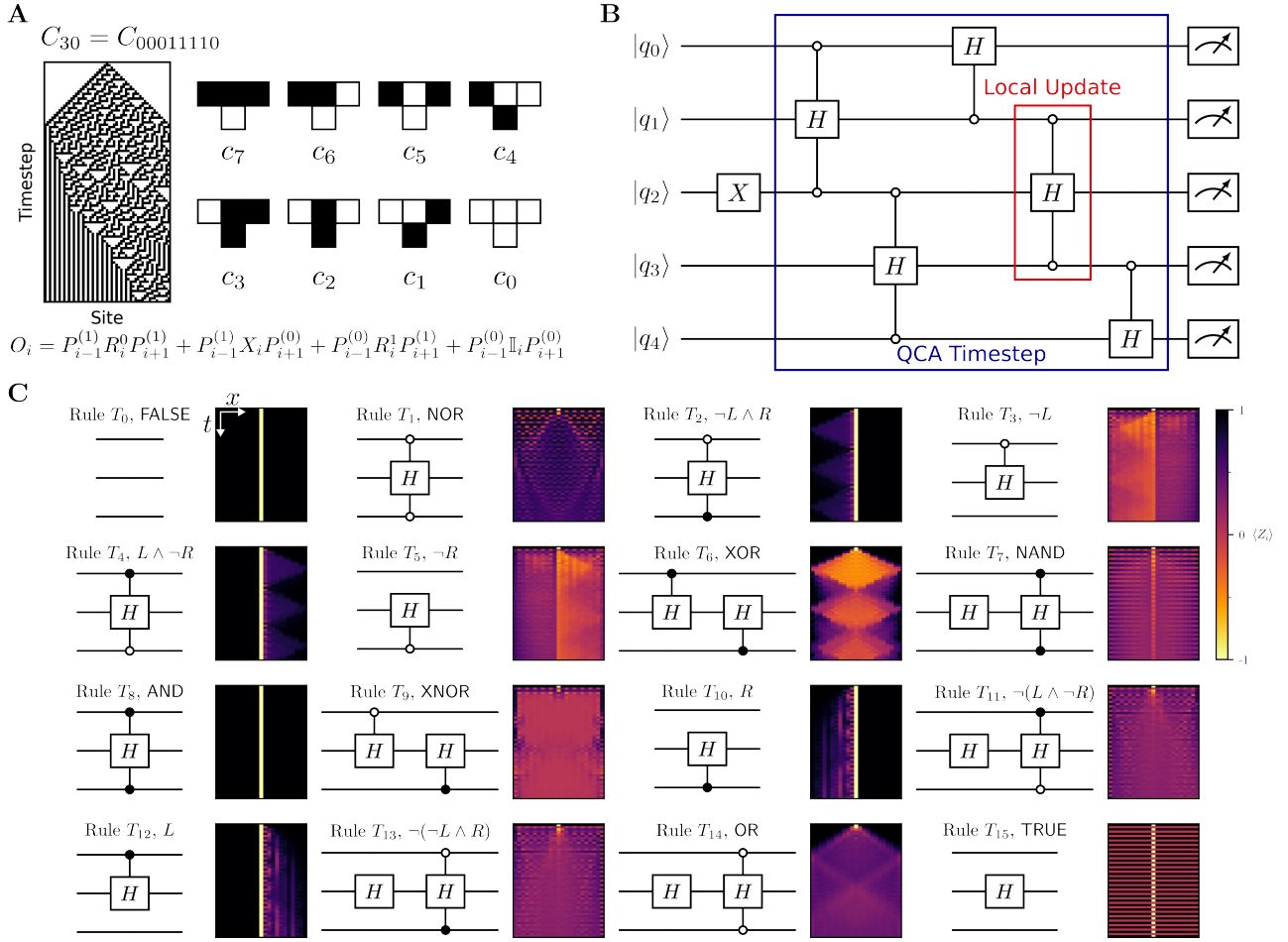


FIG. 1. **Minimal circuit blocks for unitary elementary QCA** (a) Classical spacetime evolution of ECA Rule 30, illustrating the generation of complex dynamics from strictly local updates. (b) Quantum circuit translation of C_{201} (T_1) for one timestep highlighting the even-odd partition. (c) The minimal circuit building blocks of the 16 reversible QCAs. and the corresponding spacetime evolution of $\langle Z_i \rangle$ for a central initial excitation $|00\dots010\dots00\rangle$.

cellular automata and finally into quantum circuits. For a given classical ECA C_R , the state space S^N consists of N copies of the discrete set $S = \{0, 1\}$. Let $P_i^{(0)}$ and $P_i^{(1)}$ be the projectors on state 0 or state 1 respectively at site i . We write f as an update operator O_i on the three-site neighbourhood (mn) as

$$O_i = \sum_{m,n=0}^1 P_{i-1}^{(m)} V_i^{(mn)} P_{i+1}^{(n)}. \quad (1)$$

As the classical bit at site i has two possible states, the four possible operators for each $V_i^{(mn)}$ are the identity $\mathbb{I}$, a bit flip X , a reset $R^0 =$ to 0 or a reset R^1 to 1. The rule number's coefficients c_k encode the action of the ECA on the central site for the three-site configuration k . Regrouping c_k by the (mn) configuration of sites $i-1$ and $i+1$, the coefficients $c_p = c_{4m+n}$ & $c_q = c_{4m+n+2}$ determine $V_i^{(mn)}$.

In a QCA, the chain of N classical bits is replaced by a chain of N qubits. This allows for superpositions and the

buildup of entanglement between sites. The reversible local action is not anymore restricted to a bit flip, but can be any single-qubit unitary $\hat{V}$, and is typically chosen to be an X or Hadamard gate H . In this Letter, we chose to employ the Hadamard gate H for increased entangling power and to avoid the observed features being due to peculiarities of Clifford or permutation circuits [35]. While in classical cellular automata, the update can be applied across all sites in parallel using a copy of the neighbor states, the no-cloning theorem forbids this for qubits. Instead we update the chain following an even-odd partition. We begin by mapping the 16 reversible ECA to quantum circuits and after show how to deal with the controlled resets in the other 240 irreversible ECAs.

A. Unitary QCA

From the operator representation in Eq. (1), it is straightforward to extend the ECA rule to a digital uni-

tary evolution on qubits. Let $\hat{P}_j^{(m)} = |m\rangle\langle m|$ be the projector onto the $\hat{Z}_j$ eigenstates. We renumber the 16 reversible rules following the convention established by Hillberry et al [1, 36]. Each reversible ECA, denoted as $T_{R'}$, is given a rule number $R' \in \{0, 1, \dots, 15\}$ with the binary representation of $R' = d_{11}2^3 + d_{10}2^2 + d_{01}2^1 + d_{00}2^0$ encoding whether a non-trivial action takes place depending on the different neighbourhood states. The corresponding QCA is then defined by the local update unitary

$$\hat{U}_i = \sum_{m,n=0}^1 \hat{P}_{i-1}^{(m)} \hat{V}_i^{d_{mn}} \hat{P}_{i+1}^{(n)}. \quad (2)$$

This update is applied to the wavefunction $|\psi_t\rangle$ in an even-odd partition on the qubit chain

$$|\psi_{t+1}\rangle = \prod_{i \text{ odd}} \hat{U}_i \prod_{j \text{ even}} \hat{U}_j |\psi_t\rangle. \quad (3)$$

The condition of reversibility limits us to ECAs C_R with binary coefficients respecting $c_{k+2} = 1 - c_k$ for $k \in \{0, 1, 4, 5\}$. We can then identify $d_{11} = c_7$, $d_{10} = c_6$, $d_{01} = c_3$ and $d_{00} = c_2$ to bridge the Wolfram and Hillberry numbering schemes.

While this representation can serve as the basis for simulating QCA numerically, we translate this operator representation further into quantum circuits for execution on quantum processors. In theory, each term $\hat{P}_{i-1}^{(m)} \hat{V}_i^{d_{mn}} \hat{P}_{i+1}^{(n)}$ can be written as a bi-controlled gate and each QCA locally as a sequence of at most four such commuting bi-controlled gates. In practice however, each applied gate carries a small error that adds decoherence to the system and it is essential to use the minimal number of gates involving as few qubits as possible in each layer. Since the possible neighbourhood configurations determine, whether the central qubit is updated via $\hat{V}_i$ or not, the 16 reversible ECA rules correspond to the 16 Boolean functions of two variables L (eft) and R (ight). In Fig. 2 c), we give explicit minimal circuit building blocks for each reversible ECA, making use of that fact that X and H are involutory, i.e. that they are their own inverse. The building blocks consist of at most two gates and are derived as the so-called Kronecker Reed-Muller expansions (KRME) of Boolean functions [21, 22]. We list the reversible automata and their KRME in Table I. The building blocks can be read off directly from the XOR-ed terms, which are applied successively.

B. Nonunitary QCA

The direct analogy between the operator O_i on the three-site classical state space $\{0, 1\}^3$ and operator $\hat{U}_i$ on the Hilbert space $\mathbb{C}^{2^3}$ breaks down if the reset operations R^0 and R^1 are involved. Instead, a QCA acts on the density matrix $\hat{\rho}$. Let $\hat{\sigma}^- = |0\rangle\langle 1|$ and $\hat{\sigma}^+ = |1\rangle\langle 0|$.

TABLE I. Activation logic of the 16 reversible ECAs, indexed via the Wolfram C_R or the Hilberry T_R schemes. A local update on site i is applied based on a Boolean condition on its left (L) and right (R) neighbor states. The Kronecker Reed-Muller expansion (KRME) gives the minimal circuit implementation.

C_R	$T_{R'}$	Boolean function	KRME
51	15	TRUE	1
54	14	OR	$1 \oplus \bar{L}\bar{R}$
57	13	$R \rightarrow L$	$1 \oplus \bar{L}R$
60	12	L	L
99	11	$L \rightarrow R$	$1 \oplus L\bar{R}$
102	10	R	R
105	9	XNOR	$\bar{L} \oplus R$
108	8	AND	LR
147	7	NAND	$1 \oplus LR$
150	6	XOR	$L \oplus R$
153	5	NOT R	$\bar{R}$
156	4	L AND NOT R	$L\bar{R}$
195	3	NOT L	$\bar{L}$
198	2	NOT L AND R	$\bar{L}R$
201	1	NOR	$\bar{L}\bar{R}$
204	0	FALSE	0

For continuous-time automata, the evolution can be described via a Lindblad Master equation with a set of PXP-like jump operators $\hat{L}_i^{(mn)} \propto \hat{P}_{i-1}^{(m)} \hat{\sigma}_i^\pm \hat{P}_{i+1}^{(n)}$. TODO: CITE PAPERS WHERE THIS HAS BEEN DONE. WINTERMANTEL AND MOSTLY GAVINS WORK + GUEDES. For the discrete-time QCA at hand, we describe the evolution of $\hat{\rho}$ via an even-odd application of quantum channels. These are completely positive trace preserving maps and can be decomposed into sets of Kraus operators $\{\hat{K}_\mu\}$ that are applied as $\hat{\rho}_{t+1} = \sum_\mu \hat{K}_\mu \hat{\rho}_t \hat{K}_\mu^\dagger$ and fulfill the completeness relations $\sum_\mu \hat{K}_\mu^\dagger \hat{K}_\mu = \mathbb{I}$. A reset, or decay, to $|0\rangle$ (with probability $p = 1$) on a single qubit at site i is given by the amplitude damping channel with Kraus operators $\hat{D}_{0,i} = \hat{P}_i^{(0)}$ and $\hat{D}_{1,i} = \hat{\sigma}_i^+$. Respectively, a reset, or excitation, to the state $|1\rangle$ is given by $\hat{E}_{0,i} = \hat{P}_i^{(1)}$ and $\hat{E}_{1,i} = \hat{\sigma}_i^-$. Suppose a decay of a qubit at site i occurs when the neighboring qubits are in states $|m\rangle$ and $|n\rangle$. This neighborhood-dependent decay is described by the three-site Kraus channels

$$\hat{D}_{0,i}^{(mn)} = \sum_{m',n'=0}^1 \hat{P}_{i-1}^{(m)} \left(\hat{P}^{(0)} \right)^{\delta_{mm'}\delta_{nn'}} \hat{P}_{i+1}^{(n')}, \quad (4)$$

$$\hat{D}_{1,i}^{(mn)} = \hat{P}_{i-1}^{(m)} \hat{\sigma}_i^+ \hat{P}_{i+1}^{(n)}, \quad (5)$$

with $\delta_{mm'}$ denoting the Kronecker delta.

The density matrix $\hat{\rho}_{t+1}$ describes a mixed state of different evolution pathways locally applied to $\hat{\rho}_t$ according to the probabilities $\text{Tr} \left[\hat{\rho}_t \hat{P}_{i-1}^{(m)} \hat{P}_{i+1}^{(n)} \right]$. Importantly, the evolution tracks state overlaps with the (mn) neighborhood configurations and is not conditional on a measurement of these qubits.

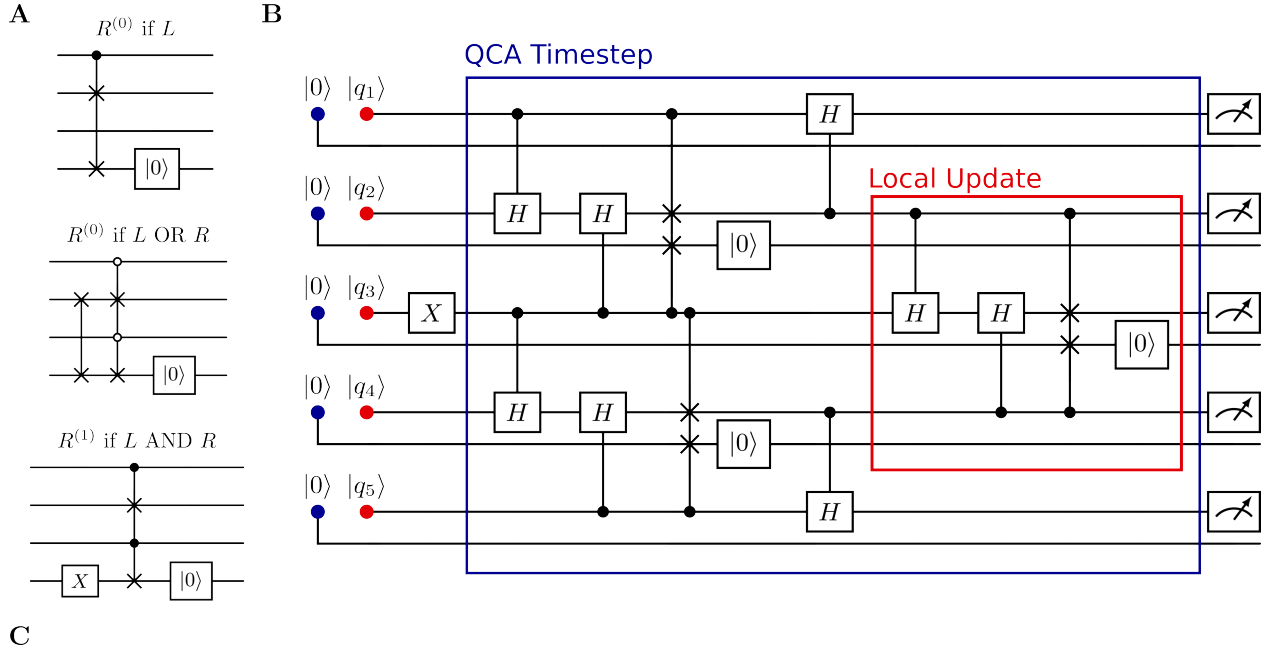


FIG. 2. **Minimal circuit blocks for non-unitary elementary QCA** (a) Three isolated circuit building blocks for different decay or excitation logic, implemented via (bi-)controlled target-ancilla SWAP gates and ancilla reset. The gate sequences are obtained from the same KRME expansions as in Table I. (b) Quantum circuit translation of the non-unitary C_{22} for one timestep using one ancilla per qubit. Alternatively, a single ancilla can be employed at the cost of long-distance operations.

In ?? a), we depict a few fundamental circuit implementations for decay and excitation channels using an ancilla qubit, that is prepared in the $|0\rangle$ state, and whose state is exchanged with the qubit at site i via a (bi-)controlled SWAP gate. Afterwards, the ancilla is destructively reset to $|0\rangle$ for use on the next qubit or in the next timestep respectively. The fundamental circuit parts follow the same KRME-based construction logic with the activation $\hat{V}_i$ unitary replaced by an ancilla $SWAP_{i,a}$. The possible updates for a qubit site consist of a unitary evolution, a decay channel or an excitation channel. We can thus associate to each of the 256 ECA, a tuple (B_U, B_D, B_E) of three Boolean functions of the two variables L (eft) and R (ight). These functions determine the activation logic of each channel and provide the fundamental building blocks that we compose into a QCA cycle, as shown in ?? b) for the irreversible ECA rule C_{22} . Because the logic functions associated to the 256 ECA are mutually exclusive and the projection-action-projection structure of each channel, the order in which they are applied commutes.

Mathematically, we show that this procedure gives the same Kraus operators as introduced before. We extend the CSWAP gate $|0\rangle\langle 0|_c \otimes \mathbb{I}_i \otimes \mathbb{I}_A + |1\rangle\langle 1|_c \otimes SWAP_{i,A}$ to two control qubits as

$$\hat{W}_i^{(mn)} = \sum_{m', n'=0}^1 \hat{P}_{i-1}^{(m')} (SWAP_{i,A})^{\delta_{mm'} \delta_{nn'}} \hat{P}_{i+1}^{(n')} \quad (6)$$

The quantum system is initialized in $\hat{\rho} \otimes |0\rangle\langle 0|_A$ and the quantum channel obtained from applying $\hat{W}_i$ and tracing

out the ancilla is described by Kraus operators

$$\hat{K}_{\mu,i}^{(mn)} = \langle \mu |_A \hat{W}_i^{(mn)} |0\rangle_A \quad (7)$$

with $\mu \in \{0, 1\}$. From $SWAP_{i,A} |\psi\rangle_i |\phi\rangle_A = |\phi\rangle_i |\psi\rangle_A$, we find $\langle 0 |_A SWAP_{i,A} |0\rangle_A = |0\rangle\langle 0|_i$ and $\langle 0 |_A SWAP_{i,A} |1\rangle_A = |0\rangle\langle 1|_i$. Since $\langle \mu |_A \mathbb{I}_i \otimes \mathbb{I}_A |0\rangle_A = \delta_{\mu 0} \mathbb{I}_i$, we recover $\hat{K}_{\mu,i}^{(mn)} = \hat{D}_{\mu,i}^{(mn)}$.

Many famously studied QCA models can easily be mapped to executable quantum circuits via the above mapping. When equipped with an X instead of a Hadamard H gate, C_{54} and C_{201} have been analytically solved. Rule 54 famously exhibits interacting solitons and rule 201 is the Floquet Rydberg PXP -model of kinetically constrained dynamics. The one-directional rules 60, 99, 153, 195 correspond to Quantum East models. In 2D, the underlying rule of the “quantum game of life” can be recast as.

TODO: ADD CITATIONS: Among these 16 QCA, rules 54, 201 and 150 have been extensively studied. Rule 54: Solitons, Rule 201: Rydberg PXP model, Rule 150: Small-world networks. One-directional models (60,99,153,195) Quantum East models. Quantum game-of-life as a bigger neighborhood, DKCA as two mixed rules.

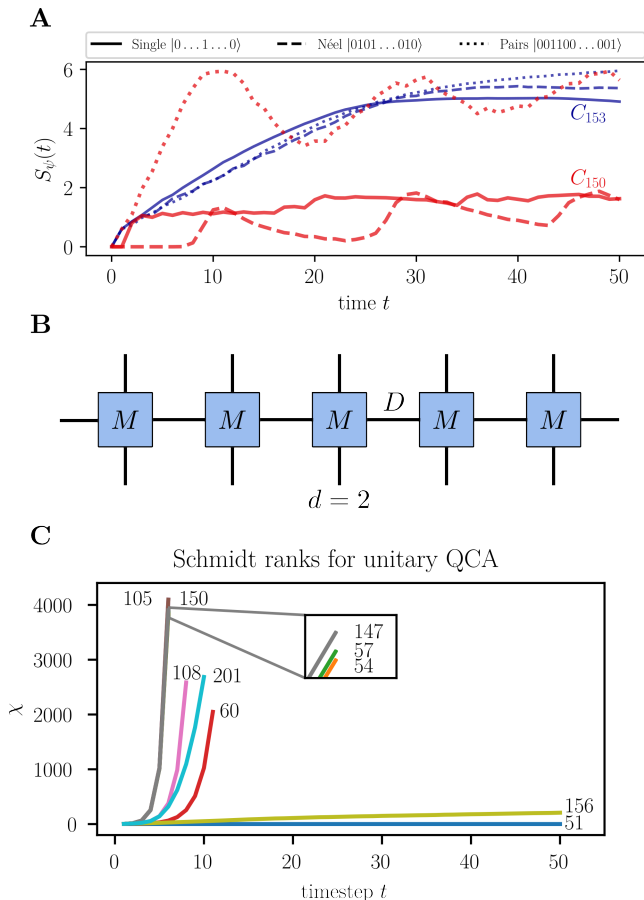


FIG. 3. **State-dependent entanglement generation and operator complexity.** (a) Bipartite half-chain entanglement entropy $S_\psi(t)$ for rules C_{150} and C_{153} initialized in single-excitation $|0\dots1\dots0\rangle$, Néel $|0101\dots010\rangle$, and pair $|001100\dots001\rangle$ states, showing C_{150} to support both area-law and volume-law dynamics. (b) Tensor network representation of the single-cycle operator $\hat{U}_C$ as a Matrix Product Operator (MPO). (c) Growth of the MPO effective Schmidt rank χ at 10^{-10} truncation over time for different QCA.

C. Hardware Implementation

IV. CLASSICAL SIMULABILITY

The boundary between classical simulability and true quantum utility is currently being outlined by a highly competitive frontier, where advanced tensor network methods continue to push past the limits of near-term quantum hardware. The classical simulability of above QCA rules is fundamentally limited by the entanglement barrier. For one-dimensional systems, Matrix Product State (MPS) algorithms propagate a quantum state $|\psi\rangle$ with $\mathcal{O}(ND^2)$ memory and $\mathcal{O}(ND^3)$ time complexity, where D is the maximum Schmidt rank, or bond dimension, of the compressed state. For the time evolution to be numerically exact, the bond dimension must scale exponentially with the bipartite entanglement entropy

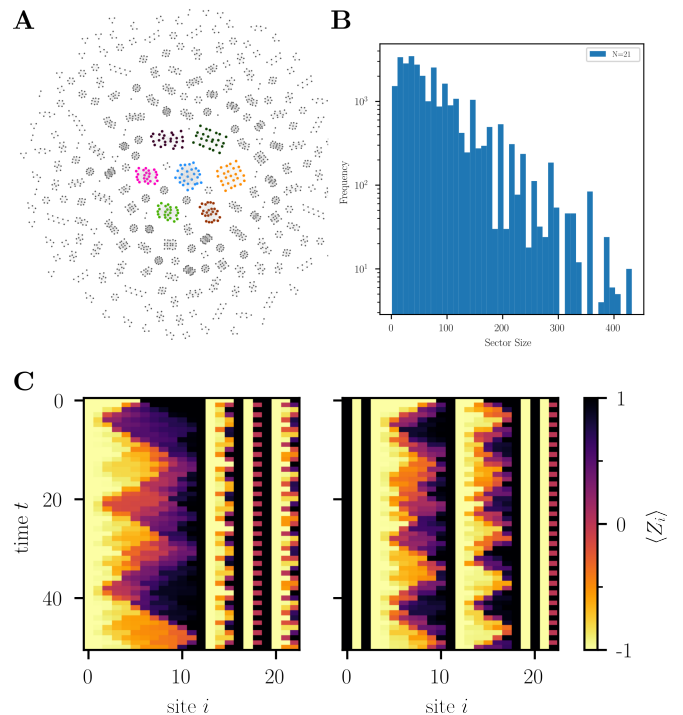


FIG. 4. **Hilbert space fragmentation in C_{156}** (a) Network graph of dynamically disconnected Krylov subsectors for the L AND NOT R rule QCA C_{156} , visualized for $N=11$. (b) Frequency distribution of sector sizes, revealing a shattering of the state space into subspaces. (c) Spatiotemporal evolution of the local magnetization $\langle Z_i \rangle$ for two distinct initial configurations, showing the $|01\rangle$ domain walls and the dynamics occurring between.

$D \geq e^{S_\psi}$. We thus probe entanglement generation in the 16 unitary QCA circuits, introduced in Section III, aiming to classify them according to their classical simulability.

While such a classification seems possible when starting from a given initial state, such as a single central excitation, we find that time evolution with many of our rules can lead both to late-time volume-law or area-law entanglement depending on the initial state. Figure 3 a) shows $S_\psi(t)$ for the two QCA C_{150} (XOR) and C_{153} (NOT R) and three different initial states. For both the single-excitation and the Néel initial states, the entanglement generated in the C_{150} automaton remains bounded to low values $S_\psi < 2$ whereas it grows fast for C_{153} and then saturates at a high value. Increasing system size reveals volume-law entanglement for C_{153} and area-law entanglement for C_{150} . A Schmidt decomposition shows that in C_{150} the maximum Schmidt rank χ is bounded for these two initial states by $\chi \leq 4$, allowing MPS simulations with hundreds to thousands of qubits. However, the pair state $|001100\dots001\rangle$ shows that this behaviour is not universal since evolution with C_{150} leads to volume-law entanglement for that state.

To characterize the dynamics by a state-independent measure, we shift our focus to the operator entanglement

entropy of the global time evolution operator itself [37–48]. Since time evolution is defined by repeated application of a fundamental cycle $\hat{U}_C$ such that $|\psi(t)\rangle = \hat{U}_C^t |\psi_0\rangle$, we study the properties of $\hat{U}_C^t$ directly. By the Choi-Jamiolowski isomorphism $\hat{U}_C(t)$ can be mapped to a state $|U_C(t)\rangle\rangle$ in the doubled Hilbert space whose Schmidt decomposition reflects the spatiotemporal evolution of quantum information spreading in the QCA [37, 47]. Numerically, we represent $\hat{U}_C$ by a Matrix Product Operator (MPO) with internal bond dimension D (see Fig. 3 b)) and obtain the MPO by direct synthesis with intermediate compressions from the corresponding quantum circuit [49]. While a QCA cycle consists of up to $2N$, possibly multi-site, gates, the resulting MPO has at most bond dimension 4 at each cut. This is because the fundamental building blocks have at most bond dimension 2 and the local even-odd updates [50]. In Fig. 3 c) we depict the growth of the maximum Schmidt rank $\chi(t)$ after SVD compression with the number t of applications.

- cite here: foundational ([37, 38, 40, 41]), opEE chaotic [44], opEE MBL [48]

TODO: What can we actually say here, contrast (at least in supplement with X models). The observed scalings correspond to a hierarchy of dynamical constraints. 105/150 no dynamical constraints, only few trivial trajectories 60 no dynamical constraints but only one-sided interactions 54,57 solitons scattering 108, 201 kinetical constraints PXP model, weak HSF 156 interactions but strong HSF (present evidence in Section IV) 51 no interactions We map the evolution operator U^t to a state $|U^t\rangle\rangle$ in a doubled Hilbert space via the Choi-Jamiolowski isomorphism, where the MPO bond dimension across a spatial cut coincides with the bipartite Schmidt rank of $|U^t\rangle\rangle$.

A. Rules 156 and 198

This linear growth is a consequence of a particularly strong Hilbert space fragmentation and emergent integrability in these two models. Patterns of the form 01, resp. 10, constitute invariant domain walls under the QCA's activation logic $L \wedge \neg R$, resp. $\neg L \wedge R$. We depict two typical spatiotemporal evolutions of the magnetization $\langle Z_i(t) \rangle$ under C_{156} with 01 domain walls and soliton-like dynamics between domain walls in Fig. 4 c). The presence of domain walls cause the Hilbert space to fragment into exponentially many, disconnected Krylov sectors. While fragmentation occurs also for C_{108} and C_{201} due to their 00 and 11 domain walls, the structure and distribution of Krylov sectors are vastly different. In the Floquet-PHP model, the largest connected Krylov sector lives within the subspace that contains zero 11 pairs, the so-called Fibonacci space, which scales asymptotically as $D_{max} \sim \phi^N$, where $\phi \approx 1.618$ is the golden ratio [26]. In contrast, the subspace with no 01 domain walls for the Floquet-PHQ model consists only of $N + 1$

states of the form $|11\dots 10\dots 00\rangle$. Thus the largest subspace must inherently contain an extensive number of domain walls. Numerically we find $D_{max} \sim 1.319^N$, which we explain by a combinatorial argument in Section B. As portrayed in Fig. 4 for $N = 21$, while the largest sector is well separated from the bulk of the distribution in the Floquet-PHP model, the sector sizes in the Floquet-PHQ model span a much narrower range without any single dominant sector. The dynamics of the Floquet-PHQ model portrayed in Fig. 4 suggest that a mapping to free non-interacting fermions within the respective intervals may be possible. The configuration of 01 domain walls partition the chain into Since any Krylov sector is uniquely defined by a static configuration of 01 domain walls, the chain is partitioned into isolated spatial intervals of lengths (l_k) . Since all states in that interval are restricted to $|11\dots 10\dots 00\rangle$ and superpositions thereof, the local Hilbert space dimension of an interval of length l_k is drastically reduced from 2^{l_k} to $l_k + 1$, and the dynamics within that interval reduce to a single particle hopping in a one-dimensional box of length l_k . There is at most one moving 10 domain wall in an interval k which can be described by a hard-core boson creation operators $b_{k,x}^\dagger$ satisfying $(b_{m,x}^\dagger)^2 = 0$. Through the standard Jordan-Wigner transformation, these hard-core bosons can be exactly mapped to spinless fermions and since the frozen 01 blockades act as infinite potential barriers, fermions in room k are strictly forbidden from scattering with fermions in adjacent rooms.

Because the $|10\rangle$ domain walls are strictly frozen, the corresponding projectors commute with the time-evolution operator, allowing the global unitary to exactly factorize over all possible static wall configurations $\{w\}$:

$$U^t = \sum_{\{w\}} \Pi_{\{w\}} \left(\bigotimes_m u_m^t \right) \Pi_{\{w\}}, \quad (8)$$

where u_m^t is the local unitary acting exclusively within the m -th isolated spatial room between walls. While free noninteracting finite-density fermion systems typically still show an exponential growth for the bond dimension $\chi(t)$ of the evolution operator MPO, the additional restriction to a single particle in each interval explains the linear $\chi(t) \propto t$ growth. When evaluating the bipartite operator entanglement across a spatial cut, the cut can intersect at most one active interval, denoted by an index c . All other intervals fall entirely to the left or right of the cut, contributing a trivial Schmidt rank of 1. Consequently, the dynamic generation of entanglement is entirely restricted to the local unitary u_c^t . For the single particle in interval c contribute to the bipartite Schmidt rank, it must cross the spatial cut. The strict locality of the QCA imposes a linear lightcone, dictating that a particle can only cross the cut if it originates within a distance $2t$ of the cut and thus bounding the nonzero elements of the transition amplitude matrix across the cut to a sub-block of maximum dimensions $2t \times 2t$. Since the Schmidt rank of a $2t \times 2t$ matrix is at most $2t$.

TODO: Last step in the inequality for the different cut configurations.

Therefore, the strong fragmentation reduces the kinetically constrained interacting model into an exactly solvable, emergent non-interacting free-fermion clockwork, strictly forbidding the exponential entanglement growth characteristic of chaotic or standard integrable systems.

B. Ergodicity breaking across models

C. Non-unitaries

MPDO approach for non-unitaries. Again even though many gates involved, low and varied bond-dimensions (show spectra?). Breakdown of HSF rule. While not sufficient to classify dynamics, MPDO evolution hints on limit cycles, fixed points, chaos.

V. CONCLUSION

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Appendix A: Superoperator Formalism

For a concise notation on the full qubit chain, we flatten the density matrix $\hat{\rho}$ into a vector $|\rho\rangle$ in the doubled space $\mathcal{H} \otimes \mathcal{H}^*$ using the Choi isomorphism $|m\rangle\langle n| \rightarrow |m\rangle|n\rangle$. We denote the projectors on the doubled space by $\mathbb{P}^{(m)} = |mm\rangle\langle mm|$. This formalism will provide us with a direct mapping of an ECA rule number R to the corresponding QCA operator. Specifically, it allows us to cast a set of Kraus operators into a single superoperator

$$\mathbb{K} = \sum_{\mu} \hat{K}_{\mu} \otimes \hat{K}_{\mu}^*, \quad (\text{A1})$$

that acts on $|\rho\rangle$ and reduces the evolution by the QCA to

$$|\rho_{t+1}\rangle = \prod_{i \text{ odd}} \mathbb{S}_i \prod_{j \text{ even}} \mathbb{S}_j |\rho_t\rangle, \quad (\text{A2})$$

where

$$\mathbb{S}_i = \sum_{m,n=0}^1 \mathbb{S}_i^{(mn)} = \sum_{m,n=0}^1 \mathbb{P}_{i-1}^{(m)} \otimes \mathbb{K}_i^{(mn)} \otimes \mathbb{P}_{i+1}^{(n)}. \quad (\text{A3})$$

We define superoperators for the unitary and nonunitary parts of the QCA's action on one qubit as

$$\mathbb{V}_i = \hat{V}_i \otimes \hat{V}_i^*, \quad (\text{A4})$$

$$\mathbb{D}_i = \sum_{\mu} \hat{D}_{\mu,i} \otimes \hat{D}_{\mu,i}^*, \quad (\text{A5})$$

$$\mathbb{E}_i = \sum_{\mu} \hat{E}_{\mu,i} \otimes \hat{E}_{\mu,i}^*. \quad (\text{A6})$$

Following application logic in ??, any of the 256 ECA is now directly mapped by its rule number $R = \sum_{k=0}^7 c_k 2^k$ to a QCA via

$$\mathbb{K}_i^{(mn)} = \mathbb{V}_i^{(1-c_p)c_q} \times \mathbb{D}_i^{(1-c_p)(1-c_q)} \times \mathbb{E}_i^{c_p c_q}, \quad (\text{A7})$$

where $c_p = c_{4m+n}$ and $c_q = c_{4m+n+2}$ are the binary expansion coefficients corresponding to the (mn) neighborhood configuration.

Appendix B: Krylov Sector Scalings

We derive the scaling law for the size of the largest Krylov sector in the Floquet-PHQ model C_{156} . The domain walls 01 shatter the Hilbert space into dynamically disconnected subsectors and each sector is uniquely defined by the position of these domain walls. Only in the intervals between domain walls, dynamics can occur. Let the N -qubit chain be partitioned into k such intervals of lengths $(l_i)_{1 \leq i \leq k}$. Since no more domain walls can be within an interval, the allowed states in each interval are of the form $11 \dots 100 \dots 0$. Thus there are each interval has a local Hilbert space dimension $d_i = l_i + 1$ and the total dimension of the Krylov sector is given by

$$D = \prod_{i=1}^k (l_i + 1). \quad (\text{B1})$$

To find the scaling of the largest possible Krylov sector we must maximize D under the constraint of a fixed chain length

$$\sum_{i=1}^k (l_i + 2) = N. \quad (\text{B2})$$

The total dimension of the Krylov sector is the sum over all valid combinatorial partitions of the chain. In the thermodynamic limit $N \rightarrow \infty$, this sum is dominated by the single maximum-entropy partition. We therefore extract the asymptotic scaling $D_{max} = b^N$ of the largest Krylov sector by applying a saddle-point approximation and assuming $l_i = l$ for all i . Under this assumption, the number of intervals becomes $k = \frac{N}{l+2}$ and $D_{max} = (l+1)^{\frac{N}{l+2}}$. Let $f(x) = \ln b = \frac{\ln x}{x+1}$ with $x = l+1$. The root of the derivative f' is given by the solution to the transcendental equation $\ln x = 1 + \frac{1}{x}$. A numerical solution gives $l = x - 1 \approx 2.5911$ and implies

that at finite N the largest sector consists of a mixture of rooms of length 2 and 3. Since $b = (l + 1)^{-(l+2)}$ has a maximum at $l = 3$, this is the strict optimal length if the total number of sites is divisible by $l + 2 = 5$.

Substituting l back into the base formula yields the

thermodynamic scaling limit

$$D_{max} = b^N \approx 1.3195^N, \quad (\text{B3})$$

confirming our numerical observation in Section IV by a combinatorial analysis. In contrast to the Floquet-PHP model C_{108} , where the largest sector size is obtained from the algebraic Fibonacci sequence, the largest sector size for C_{156} is obtained from the solution of a transcendental saddle point equation.

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